Detecting and quantifying spatial information leakage in predictive modelling: a diagnostic framework and the `spLeakage` R package

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June 23, 2026

Abstract

Spatial predictive models—in disease mapping, species distribution modelling, digital soil mapping and remote sensing—routinely report accuracy estimated by random cross-validation (CV). When observations are spatially autocorrelated, training and test points that fall close together share information, so the reported error measures near-interpolation rather than the out-of-sample prediction the map is actually used for. The resulting accuracy is *optimistically biased*. A family of spatially corrected CV schemes exists to *prevent* this, but they *generate folds*; none takes a split a user has *already* made and reports how much it leaks or how inflated the numbers are. The field is moreover split by a live controversy: nearest-neighbour-matching CV (NNDM/kNNDM) argues CV geometry must imitate prediction geometry, whereas design-based critics argue that for probability samples random CV is unbiased and spatial CV is *pessimistic*. We resolve both problems with a single thesis: **spatial leakage and optimism are only well-posed relative to a declared design $\times$ estimand $\times$ target**, and the same split can be leaking, correct, or pessimistic depending on those declarations. Under this framing we (i) give optimism a formal estimand—excess explained variance under a Gaussian process—together with a closed-form nearest-neighbour expression and a Wasserstein-1 bound; (ii) define a *signed* Spatial Leakage Index (SLI) that audits any split post-hoc and decomposes leakage to folds and points; (iii) quantify optimism through a refit-free, simulation-calibrated emulator that carries its own area-of-applicability guard; and (iv) turn the controversy into an estimand-first recommendation engine. Across exact Gaussian-process algebra the SLI is a near-sufficient statistic for true optimism ($r = 0.98$), the signed index predicts *signed* optimism where NNDM's unsigned statistic cannot ($r = 0.98$ vs -0.09), and the emulator attains held-out $R^2 \approx 0.76$ with $\approx 94\%$ interval coverage. A 16-dataset meta-audit finds reported accuracy optimistic in **100%** of cases (median 22%), and reveals that optimism decomposes into an autocorrelation channel (which spatial CV diagnoses) and a trend-extrapolation channel (which it misses). All methods are implemented in the open-source `spLeakage` R package, which audits rather than generates folds and interoperates with the established spatial-CV ecosystem.

Keywords: spatial cross-validation; map accuracy; information leakage; optimism; area of applicability; Gaussian process; distribution shift; disease mapping; species distribution models; reproducibility.

1 Introduction

Maps produced by statistical and machine-learning models underpin decisions in public health, conservation, agriculture and earth observation. The credibility of such a map rests almost entirely on a single reported number—its estimated predictive accuracy. Yet for spatial data this number is systematically unreliable. Observations collected over space are autocorrelated: nearby locations have similar values. When a model is validated by an ordinary random train/test split, test points routinely have a training neighbour only a short distance away. The model is therefore evaluated on a task close to *interpolation*, whereas at deployment it must predict at locations far from any training data. The reported error understates the true error: it is *optimistically biased*, an instance of *information leakage* from training to test (Roberts et al., 2017; Ploton et al., 2020; Kattenborn et al., 2022).

The consequences are not academic. Ploton et al. (2020) showed in *Nature Communications* that large-scale ecological maps are far less accurate than reported once spatial structure is respected. The same overconfidence has been documented for convolutional networks on remote-sensing imagery (Kattenborn et al., 2022) and is a recurring concern in species distribution modelling and digital soil mapping. A decision-maker who trusts an inflated accuracy figure will over-rely on a map that is, in the regions that matter most (those far from data), much weaker than advertised.

The methodological response, and its gap. A mature family of *spatially aware* cross-validation methods exists to avoid the problem by constructing better folds: spatial block CV (Roberts et al., 2017; Valavi et al., 2019), buffered / spatial leave-one-out CV (Le Rest et al., 2014; Telford and Birks, 2009), and nearest-neighbour distance matching CV (NNDM/kNNDM; Milà et al., 2022; Linnenbrink et al., 2024), packaged in tools such as `CAST` (Meyer et al., 2024), `blockCV` (Valavi et al., 2019), `spatialsample` and `sperrorest` (Brenning, 2012). These are *fold generators*. They answer "how should I cross-validate?". None of them answers the question a practitioner who has already run an analysis actually asks: *"I already have a split (or a published result). How much does it leak, and how inflated is my reported accuracy?"* There is no leakage *diagnostic*, no *audit*, and no *optimism estimate* for an arbitrary split. This is the gap `spLeakage` fills.

A live controversy. Worse, the field disagrees about what "correct" validation even is. Milà et al. (2022) and Linnenbrink et al. (2024) argue that the CV geometry should match the prediction geometry, making random CV optimistic for clustered data. Wadoux et al. (2021) argue, from a design-based standpoint, that "spatial cross-validation is not the right way to evaluate map accuracy": for a *probability sample*, random validation is design-unbiased and spatial CV introduces a *pessimistic* bias (see also de Bruin et al., 2022). **Both camps are right—in different regimes.** A practitioner has no tool that tells them which regime they are in, and therefore which validation is correct.

Our contribution. We argue that the gap and the controversy have a common root and a common resolution. Spatial leakage and optimism are *not* properties of a dataset or a split in isolation; they are only well-defined *relative to a declared sampling design, a declared estimand, and a declared prediction target* (Section 2). The same split can be leaking, correct, or pessimistic according to these three declarations. Making them explicit (i) dissolves the Milà–Wadoux controversy—each side is correct in a different cell of the design × estimand × target space—and

(ii) removes a circular dependency that would otherwise undermine any optimism estimate. On this foundation we build four contributions:

1. **A formal estimand and theory for optimism** (Section 3). Under a Gaussian process, optimism equals the *excess explained variance* a CV scheme enjoys over deployment; this yields a closed-form nearest-neighbour expression and a Wasserstein-1 bound that connect spatial leakage to distribution-shift generalization theory.
2. **A signed Spatial Leakage Index (SLI)** (Section 4): a post-hoc, dependence-aware, *signed* leakage score for any split, with a decomposition to folds and points (a leakage map) that fold-generators cannot produce.
3. **A refit-free optimism estimator** (Section 5): an empirical route and a simulation-calibrated emulator with adaptive predictive intervals and its own area-of-applicability guard—"your reported error is $X\%$ optimistic (a–$b\%$)".
4. **A design-aware recommendation and correction layer** (Section 6): an estimand-first decision engine, a de-leaked accuracy estimate that can correct a *published* number without refitting, and an optimal design for *validation* survey placement.

Every method is implemented in the open-source `spLeakage` R package (Section 7), which *audits rather than generates* folds and interoperates with `CAST`, `blockCV`, `spatialsample` and `mlr3spatiotempcv`. Section 8 validates every claim with exact Gaussian-process algebra, a head-to-head benchmark against NNDM, an out-of-sample emulator study, independent real-data validation, a 16-dataset meta-audit, and a malaria case study. Section 9 discusses scope, limitations and the new two-channel decomposition of optimism that the meta-audit reveals.

2 Leakage is well-posed only relative to design, estimand and target

The central claim of the paper is a reframing:

> *"Spatial leakage" and "optimism" are not properties of a dataset or a split in isolation. They are well-defined only relative to a declared sampling* ***design****, a declared* ***estimand****, and a declared prediction* ***target****. The same split can be leaking, correct, or pessimistic depending on these three declarations.*

The three declarations play distinct roles.

Sampling design. Whether the data are a probability sample (with known inclusion probabilities), a designed cluster sample, or a convenience sample decides whether random validation is unbiased (Wadoux et al., 2021; de Bruin et al., 2022) or optimistic (Milà et al., 2022). Crucially, design-basedness is a fact about *how the data were collected*; it cannot be recovered from the coordinates. A designed cluster sample and a convenience sample can produce identical point patterns. Geometry (Clark–Evans index, Ripley's K) can therefore only *flag risk*; the design is an *elicited input*, never an inference.

Estimand. *Population-mean map accuracy over a region* (a design-based quantity) and *conditional predictive skill at a location* (a model-based quantity) are different questions. Conflating them is a common error and a frequent reviewer objection. We separate them and never recommend a validation scheme *across* estimands.

Prediction target. Within-sample interpolation, wall-to-wall mapping, and new-region extrapolation imply different deployment geometries, and hence different "correct" validation geometries.

This reframing is what makes "optimism" computable. We define optimism not against NNDM-as-truth, but against *the validation that is correct for the declared design × estimand × target.* For a probability sample with a population-mean estimand, the correct baseline is design-based/random validation, so a spatially blocked scheme registers as *pessimism* (negative optimism)— exactly the Wadoux case, now reported as such rather than hidden. The declarations are inputs, not inferences, which is what removes the circularity that would otherwise make the optimism estimate ill-posed. The reconciliation of the controversy is, in this sense, the paper's primary intellectual contribution; the index, the emulator and the recommender are its instantiations.

3 Theory: optimism as a distribution-shift gap

We now give optimism a formal estimand and show that a simple geometric index is its leading term. This converts the SLI from a validated heuristic into a principled method and connects spatial leakage to distribution-shift theory.

3.1 Model and the explained-variance estimand

Let Z be a zero-mean Gaussian process with stationary covariance $\mathrm{Cov}(Z(s), Z(s')) = \sigma^2\rho(\|s - s'\|)$, observed with independent nugget noise,

$$Y(s) = \mu + Z(s) + \varepsilon(s), \qquad \varepsilon \sim N(0, \tau^2), \quad \varepsilon \perp Z. \tag{1}$$

Write the total variance $V = \sigma^2 + \tau^2$ and the *signal proportion* $w = \sigma^2/V \in [0, 1]$. Simple kriging of $Y(s_0)$ from a training set T of m observations has mean-squared error

$$\mathrm{MSE}(s_0 \mid T) = V - c_0^\top \Sigma^{-1} c_0, \qquad c_0 = \sigma^2\rho(s_0, T), \quad \Sigma = \sigma^2 R + \tau^2 I, \tag{2}$$

where R is the correlation matrix of T. Call $\mathrm{EV}(s_0 \mid T) = c_0^\top \Sigma^{-1} c_0 \geq 0$ the *explained variance*: the reduction in predictive variance the training data contributes, large when s_0 is near (correlated with) the training points, small when far.

A CV scheme predicts each test point t from its fold's training set $\mathrm{Tr}(t)$; deployment predicts each target location p from the full sample S. Averaging (2), the constant V cancels and the optimism—the gap between the true deployment error E_{true} and the cross-validated error E_{cv}—is

$$\text{optimism} = E_{\text{true}} - E_{\text{cv}} = \underbrace{\tfrac{1}{n}\textstyle\sum_t \mathrm{EV}(t \mid \mathrm{Tr}(t))}_{\text{CV explains}} - \underbrace{\tfrac{1}{m}\textstyle\sum_p \mathrm{EV}(p \mid S)}_{\text{deployment explains}}. \tag{3}$$

Proposition 1 (Optimism is excess explained variance)**.** *Equation* (3) *holds for* any *Gaussian process, with no approximation. Optimism is exactly the excess explained variance that the CV scheme enjoys over deployment.*

This is the formal estimand: an interpretable population quantity, not a heuristic. It also makes the sign meaningful—optimism is positive when CV is "easier" than deployment and negative (pessimism) when CV is harder, the Wadoux regime.

3.2 The nearest-neighbour closed form and where the SLI appears

Keeping only the nearest training point, at distance d, gives $c_0 = \sigma^2\rho(d)$, $\Sigma = V$, and hence the pointwise loss

$$L(d) = \mathrm{MSE} \approx V\,[\,1 - w^2\rho(d)^2\,]. \tag{4}$$

Substituting into (3), with g the test→train and f the deployment→sample nearest-neighbour distances,

$$\text{optimism} \approx V\,w^2\Big(\mathbb{E}_t[\rho(g)^2] - \mathbb{E}_p[\rho(f)^2]\Big). \tag{5}$$

Equation (5) is, up to the scale Vw^2, exactly a difference of mean retained spatial correlation between CV and deployment—the Spatial Leakage Index of Section 4. The theory therefore predicts *which* features govern optimism: the correlation function ρ (hence range and smoothness), the signal proportion w, and the nearest-neighbour-distance laws of the split and the target. These are precisely the inputs of the emulator (Section 5); the emulator was empirically rediscovering (5).

3.3 A Wasserstein bound

Let P_{cv} and P_{dep} be the laws of the test→train and deployment→sample nearest-neighbour distances. By (4),

$$\text{optimism} \approx \int L\, d(P_{\text{dep}} - P_{\text{cv}}), \tag{6}$$

a distribution-shift generalization gap: the same loss L under two laws of the governing distance. Because L is bounded-Lipschitz, the Kantorovich–Rubinstein duality (Villani, 2009) gives

$$|\text{optimism}| \le \|L'\|_\infty\, W_1(P_{\text{cv}}, P_{\text{dep}}), \tag{7}$$

where W_1 is the Wasserstein-1 distance between the two ECDFs—which is, up to the range normalisation, the distance-form index SLI_d. Thus SLI_ρ gives the optimism *value* (5) and SLI_d gives a rigorous *bound* (7). This framing also imports the distribution-shift toolkit (importance weighting, integral-probability-metric bounds) as a route to model-agnostic optimism bounds beyond the Gaussian case. Scope and the multi-neighbour correction are discussed in Section 9; numerical validation is in Section 8.1.

4 The signed Spatial Leakage Index

4.1 Construction

For a split, let $g_{\text{obs}}(t) = \min_{s\in\text{Tr}(t)} d(t,s)$ be the test→train nearest-neighbour distance and $f_{\text{pred}}(p) = \min_{s\in S} d(p,s)$ the deployment→sample distance, where d is geodesic for geographic coordinate reference systems and projected otherwise. Their ECDFs are $\widehat{G}_{\text{obs}}$ and $\widehat{G}_{\text{pred}}$. Leakage means test points are *closer* to their training data during CV than deployment points are to the sample, i.e. $\widehat{G}_{\text{obs}}$ is shifted to smaller distances than $\widehat{G}_{\text{pred}}$.

Distance form. The signed area between the ECDFs,

$$A = \int_0^\infty (\widehat{G}_{\text{obs}}(r) - \widehat{G}_{\text{pred}}(r))dr = \mathbb{E}_p[f_{\text{pred}}] - \mathbb{E}_t[g_{\text{obs}}], \tag{8}$$

is literally how much farther deployment reaches than CV pretended, in distance units. Normalising by the variogram practical range φ (the intrinsic dependence length, *not* the map extent, which gives cross-dataset invariance) yields $\text{SLI}_d = A/\varphi$. To prevent cancellation when the ECDFs cross (leakage at short range, pessimism at long range) we report the crossing decomposition $A_+ = \int \max(\widehat{G}_{\text{obs}} - \widehat{G}_{\text{pred}}, 0)$, $A_- = \int \max(\widehat{G}_{\text{pred}} - \widehat{G}_{\text{obs}}, 0)$, $A = A_+ - A_-$, and $W = A_+ + A_-$. Here W is exactly the unsigned NNDM/kNNDM Wasserstein statistic, so SLI_d reduces to W in magnitude when the ECDFs do not cross—the explicit bridge to the established literature. The directionality ratio $\delta = A/W \in [-1,1]$ flags whether a single number is trustworthy ($\delta \approx \pm 1$) or the user should inspect the ECDF plot ($\delta \approx 0$).

Dependence form (headline). Because geographic distance is not statistical dependence, we replace raw distances by the fitted spatial correlation $\rho(h)$ and define the retained spatial information

$$c_{\text{obs}} = \tfrac{1}{n}\textstyle\sum_t \rho(g_{\text{obs}}(t)), \quad c_{\text{pred}} = \tfrac{1}{m}\textstyle\sum_p \rho(f_{\text{pred}}(p)), \quad \boxed{\text{SLI}_\rho = c_{\text{obs}} - c_{\text{pred}} \in [-1,1].} \tag{9}$$

SLI_ρ is bounded, dimensionless and signed: + means CV retains more spatial information than deployment has (optimistic leakage), − means pessimism. Under anisotropy or non-stationarity a directional/local ρ is used, so the index tracks *dependence* rather than geometry. By Section 3, SLI_ρ is the leading term of true optimism (scaled by Vw^2), which is what validates the index.

4.2 Decomposition: the leakage map

The per-observation contribution $\ell(t) = \rho(g_{\text{obs}}(t)) - c_{\text{pred}}$ satisfies $\text{SLI}_\rho = \frac{1}{n}\sum_t \ell(t)$. Mapping $\ell(t)$ over space shows *where* a split leaks; averaging within folds shows *which* folds leak. This per-point/per-fold decomposition of an arbitrary split is the substantive contribution of the index—it is exactly what a fold-generator cannot produce.

4.3 Why this is not "NNDM's W renamed"

The contribution is the *signing*, the *decomposition*, and the *post-hoc audit of an arbitrary split*, not the underlying ECDF comparison. NNDM constructs folds to minimise W; the SLI takes any split and reports a signed, dependence-aware, mappable diagnostic. Section 8.2 shows quantitatively that the unsigned W cannot recover the sign of optimism, whereas the signed SLI can.

4.4 Uncertainty

φ and ρ are estimated, often from the very clustered data that leaks. The default is a plug-in SLI from a fitted variogram; optionally, variogram parameters are sampled from their uncertainty and the SLI is recomputed per draw, giving a Monte-Carlo interval whose coverage is checked in simulation.

5 Quantifying optimism

The SLI says *whether* and *in which direction* a split leaks. The practitioner wants the number in their own currency: "my reported RMSE is $X\%$ optimistic". We provide two routes.

(a) Empirical route. Re-evaluate the user's model under the *design-matched correct* scheme (block or buffered-block CV for the clustered case) and report the gap to the user's scheme. This is accurate but model-conditional and requires refits; it is reported as model-conditional and can be negative (pessimism). It is implemented model-agnostically via a pluggable prediction function (inverse-distance weighting by default).

(b) Emulator route (refit-free). A pre-trained predictor maps (range, smoothness, signal w, sample size n, expected optimism, calibrated once by a large simulation study shipped with the package and applied instantly with no refitting. The generator is deliberately rich: Matérn fields across smoothness $\{0.5, 1.5, 2.5\}\times$ signal $\times$ {random, clustered (varied tightness), preferential (Diggle et al., 2010)} sampling $\times$ $n \in \{150, 300, 600\}$ $\times$ {Gaussian, Poisson, Binomial} responses $\times$ {IDW, random forest, GAM} learners $\times$ {grid, interpolation} targets. The point estimate is a gradient-boosted tree ensemble (Chen and Guestrin, 2016); predictive intervals are produced by *normalised split-conformal* regression (Vovk et al., 2005; Romano et al., 2019), grouped by configuration so that interval width scales with a local difficulty estimate (the mean residual of feature-space neighbours). Critically, the emulator carries its *own* area-of-applicability check (Meyer and Pebesma, 2021): queries outside the training envelope are flagged and the estimate is withheld—the tool must not commit the sin it polices. Section 8.3 reports its out-of-sample performance and its correct refusal of an out-of-distribution real-data query.

6 Recommendation, correction and validation design

6.1 Estimand-first recommendation

The recommendation engine turns the thesis into actionable, *conditional* guidance. It elicits the estimand first (population-mean map accuracy vs conditional predictive skill), then recommends over design $\times$ target (Table 1). Three rules keep it defensible: design-basedness is *elicited, never inferred from geometry*; every cell is backed by a citation or a simulation row; and guidance

Table 1: The design × target recommendation matrix (after the estimand is declared). Each cell is backed by a citation or a simulation-study row; the design is an elicited input, never inferred from the point pattern.

Sampling design \ target	**Within-sample interpolation**	**Wall-to-wall mapping**	**New-region extrapolation**
Probability / design-based	Random CV (design-based)	Design-based estimator	Caution; AOA
Clustered / preferential	NNDM / buffered	NNDM / kNNDM	Block + AOA
Convenience / opportunistic	Buffered LOO	kNNDM	Block + AOA, flag risk

is phrased conditionally ("*if* your estimand is X under framework Y, *then*...."), surfacing the assumption rather than silently adopting a side. Uncovered designs (spatiotemporal, citizen-science, preferential intensity, multi-source fusion) are flagged and referred, not forced into the table. Loudly, it also states *when spatial CV is not appropriate*—the nuance no current tool surfaces. A data-driven companion, `rank_cv_schemes()`, builds each candidate scheme, measures its SLI_ρ against the declared target, and ranks by deployment match.

6.2 The de-leaked estimator

`deleak_estimate()` corrects a *reported* accuracy value without refitting the model, using a fold-bootstrap confidence interval. A `target=` argument selects the control scheme whose SLI_ρ against the declared target is closest to zero (NNDM matching as a selection rule). This makes `spLeakage` a *meta-audit* instrument: it can correct a published number given only the data geometry and the reported metric (Section 8.5).

6.3 Optimal design for validation

Inverting the SLI yields a survey-design tool. Given training locations, a deployment target and a budget of n_v independent validation points, `design_validation()` stratifies by distance-to-training on the deployment distribution, models the per-stratum error from the GP loss $L(d)$ (4), and allocates the budget by Neyman allocation with Horvitz–Thompson inclusion weights so the weighted-mean accuracy estimate stays unbiased. This is optimal experimental design for *validation* (not prediction), which the spatial-CV literature does not address (Section 8.7).

7 The spLeakage package

`spLeakage` is an R ($\geq$ 4.1) package built on `sf` (Pebesma, 2018) with all heavy adaptors (`rsample`, `mlr3`, `caret`, `blockCV`, `terra`, `gstat`) in `Suggests` and guarded at run time. It *audits rather than generates* folds, interoperating with the spatial-CV ecosystem rather than competing with it. The core functions are: `prediction_target()` to declare deployment; `detect_leakage()` for the signed SLI; `estimate_optimism()` and `predict_optimism()` for the two optimism routes; `recommend_validation()` and `rank_cv_schemes()` for guidance; `deleak_estimate()` for correction; `design_validation()` for survey design; and per-channel detectors for grouped/duplicated-location, feature-space, temporal and trend leakage, each paired with a fix (`group_kfold()`, `temporal_kfold()`, design-matched controls). `audit_workflow()` inspects an existing resampling object and `report_leakage()` produces a submission-ready scorecard. All constructors return classed S3 objects with `print`/`summary`/`plot` methods. The shipped emulator is serialised compactly ($\approx 85\,\mathrm{KB}$) inside the package. `R CMD check` passes cleanly (0 errors/warnings/notes beyond the standard new-submission note).

```
library(spLeakage)

# 1. Declare where the model will actually be used (a wall-to-wall map)
tgt <- prediction_target(grid = pred_grid, type = "grid")

# 2. Diagnose an existing split: signed index + uncertainty + decomposition
lk <- detect_leakage(data, split = my_folds, target = tgt,
                     response = "y", n_boot = 300)
lk            # SLI_rho, 90% CI, NNDM W, signed verdict
plot(lk)      # observed-vs-deployment NN-distance ECDFs and the leakage map

# 3. Quantify the optimism it causes (refit-free emulator + empirical route)
predict_optimism(lk)
estimate_optimism(data, split = my_folds, response = "y", control = "block")

# 4. Get design-aware advice (design is declared, not inferred)
recommend_validation(data, estimand = "prediction",
                     design = "clustered", target = "grid")

# 5. Correct a reported metric without refitting
deleak_estimate(reported_rmse = 0.42, data = data,
                split = my_folds, target = tgt)
```

Listing 1: A complete audit-quantify-recommend workflow.

8 Results

We validate each claim in turn: the theory (exact GP algebra), the benchmark against NNDM, the emulator (out-of-sample), independent real-data validation, the meta-audit, the malaria case study, and the validation-design tool. Scripts to reproduce every result ship in the package's `data-raw/` directory.

8.1 The SLI is a near-sufficient statistic for optimism

Across 192 Gaussian-process configurations (range $\times$ signal $\times$ n $\times$ {random, clustered}), with explained variance computed *exactly* from the covariance algebra (no field realisation), the single-nearest-neighbour closed form (5) tracks exact optimism at $r = 0.975$ (Table 2). The package's unsquared SLI_ρ correlates with exact optimism at $r = 0.976$—explaining $\approx 95\%$ of the variance—and *beats* the single-NN squared form ($r = 0.841$), because with realistic dense training the explained variance saturates and the linear correlation difference is the better sufficient statistic. The Wasserstein bound (7) holds in 100% of configurations for the single-NN optimism

Table 2: Exact-GP validation of the theory (192 configurations).

Claim	Result
Single-NN closed form (5) predicts exact optimism	$r = 0.975$
SLI_ρ is a near-sufficient statistic for GP optimism	$r = 0.976$
Unsquared SLI_ρ beats single-NN squared form	0.976 vs 0.841
Wasserstein bound (7) holds (single-NN optimism)	100%
Multi-NN amplification (exact / single-NN)	median 1.56

Table 3: Predicting exact GP optimism from the signed SLI vs NNDM's W (288 configurations, 64% pessimistic).

Task	signed SLI_ρ	NNDM W (unsigned)
Predict *signed* optimism	$r = +0.977$	$r = -0.090$
Predict magnitude |optimism|	$r = 0.955$	$r = 0.672$
Direction (sign) agreement	97%	silent by construction

it governs; the exact multi-neighbour optimism is $\approx 1.6\times$ larger, so bounding it requires inflating $\|L'\|$ by that factor (an honest, quantified caveat). A k-neighbour extension improves magnitude but reduces correlation (noisy-OR over-counts correlated neighbours), so $k = 1$ is retained as default—a clean negative result that vindicates the original choice.

8.2 Signed SLI vs NNDM's unsigned W

The natural objection—"isn't this just NNDM's W?"—is answered against exact GP optimism over 288 configurations spanning design $\times$ split $\times$ target $\times$ range $\times$ signal, of which 64% are *pessimistic* (the Wadoux regime), so the test is not contrived. The signed SLI_ρ predicts *signed* optimism at $r = 0.977$; NNDM's unsigned W at $r = -0.090$ (Table 3). W cannot sign the bias by construction: a practitioner with a large W does not know whether to trust their numbers more or less. The SLI recovers the direction in 97% of cases, and even on magnitude beats W (0.955 vs 0.672) because W is pure geometry while SLI_ρ carries the signal proportion w that scales optimism. NNDM solves a fold-construction problem (make W small); `spLeakage` solves a diagnosis problem (how much, and which way, is accuracy biased).

8.3 The emulator generalises out of sample

On a paper-scale study (1000 configurations $\times$ 6 realisations, ≈ 2000 denoised rows), held out *by configuration*, the emulator attains $R^2 \approx 0.76$, places 97% of held-out queries inside its area-of-applicability, and its 90% predictive intervals cover at $\approx 94\%$ (Figure 1). The empirical correlation $\mathrm{cor}(\mathrm{SLI}_\rho, \text{optimism}) = 0.67$ provides direct evidence for the SLI $\to$ optimism mechanism (Figure 2). The design effect that reconciles the controversy reproduces in the data: clustered/preferential sampling makes random CV more optimistic than a probability-like random sample, while for well-covered grid targets random CV is fine or slightly pessimistic (Figure 3). The *ordering* is robust; the absolute level depends on target geometry and model class, which the emulator's model-class factor absorbs.

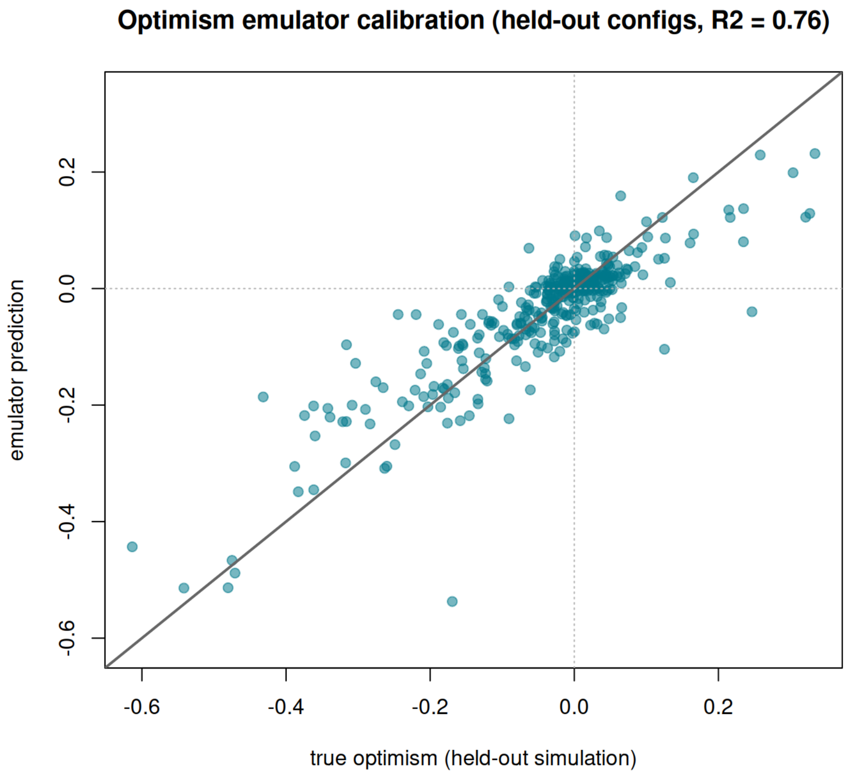

Figure 1: Out-of-sample calibration of the optimism emulator (held out by configuration). Predicted vs true optimism with conformal predictive intervals; held-out $R^2 \approx 0.76$, 90% interval coverage $\approx$ 94%, 97% of queries inside the area of applicability.

8.4 Validation against true independent error

Simulations could be a toy, so we test on three real spatial fields (meuse log-zinc, Paraná rainfall, ca20 calcium; 240 replicates), each trained on part of the data and evaluated on a *genuinely independent* held-out set, under both extrapolation (hold out a coherent region) and interpolation (hold out an interspersed subset) deployments. Under extrapolation, naive random 10-fold CV *understates the true independent error by a median of* 36%: the problem is real and large. The target-aware SLI separates the regimes as real optimism does (mean SLI +0.28 extrapolation vs -0.01 interpolation; mean real optimism +0.35 vs -0.04; $\mathrm{cor}(\mathrm{SLI}_\rho, \text{real optimism}) = 0.52$ across all replicates—strong for real, noisy $n \approx 150$ data). The de-leaked estimate recovers 58% of the gap with a fixed block control and 71% with the target-aware control, reaching $0.89\times$ true independent error without ever seeing the held-out data. The residual 29% is the hardest extrapolation, where the honest answer is area-of-applicability abstention rather than a number.

8.5 Meta-audit: how overconfident is the field?

The impact question (cf. Ploton et al., 2020): is reported accuracy systematically optimistic across real datasets? Using `spLeakage` as the instrument on 16 real spatial response variables from eight public datasets (heavy metals, soil chemistry, rainfall, elevation, ozone, malaria), with real optimism measured by spatial region-holdout, we find reported accuracy optimistic in

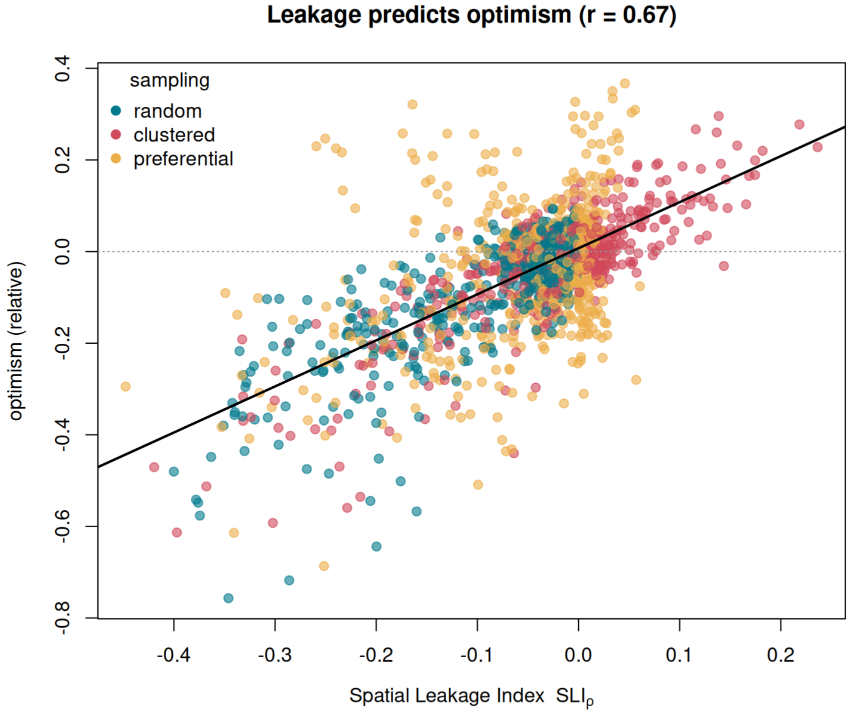

Figure 2: The Spatial Leakage Index tracks true optimism. Across the simulation grid, SLI$_\rho$ increases monotonically with measured optimism (cor = 0.67 empirically; $r = 0.98$ against exact GP optimism), the empirical counterpart of the closed form (5).

16/16 (100%) of cases, with median real optimism 22% (IQR 17–34%) and $> 20\%$ in two-thirds (Figure 4).

The deeper, honest finding is that optimism has *two* channels. The raw SLI$_\rho$ does *not* transfer across datasets (cor $= -0.39$)—which is the theory, not a bug: optimism scales as $Vw^2 \cdot (\cdots)$, so cross-dataset comparison needs the signal variance the bare index omits (the SLI is a *within*-dataset predictor, validated at cor = 0.52 above). But even a de-leaked estimate transfers only weakly (+0.23), because the smooth, trend-dominated fields (rainfall, elevation, ozone) have the *highest* real optimism (40–54%) yet near-zero short-range leakage: their error is the model's inability to extrapolate a large-scale *trend*, to which autocorrelation-based diagnostics (NNDM, the SLI, the de-leak) are blind by construction. Adding a trend-strength feature (variance explained by a quadratic coordinate surface) recovers exactly that signal (cor = +0.75, $R^2 = 0.56$ alone; $R^2 = 0.62$ with $|\mathrm{SLI}_\rho|$). Real-data optimism is therefore a two-channel phenomenon—autocorrelation (diagnosed by `detect_leakage`) and trend extrapolation (by `trend_strength`)—and the 22% median is a *lower bound* on what autocorrelation-aware tooling alone would catch.

8.6 Case study: malaria prevalence in Nigeria

On Malaria Atlas Project *Plasmodium falciparum* prevalence (Weiss et al., 2019) ($n = 66$ geolocated surveys), a naive random 10-fold split gives SLI$_\rho$ = +0.212 despite a near-zero variogram signal proportion—an apparent contradiction the package resolves. Deduplicating

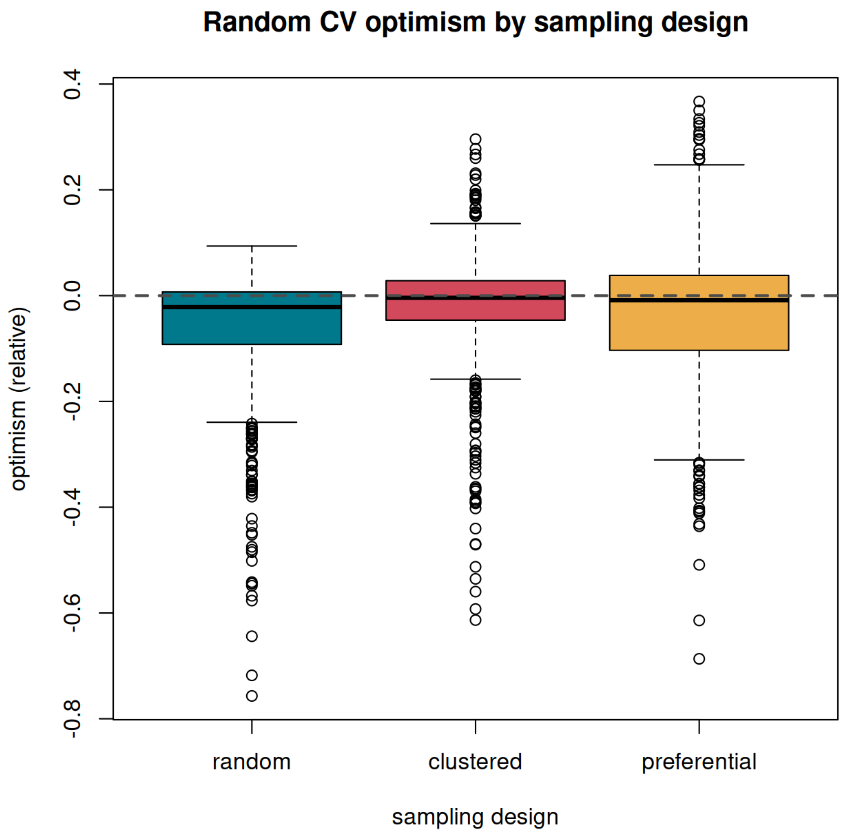

Figure 3: Reconciling Milà vs Wadoux. The optimism of random CV depends on sampling design and target: clustered/preferential designs shift random CV toward optimism (the Milà regime), whereas for probability-like samples on well-covered targets random CV is unbiased or slightly pessimistic (the Wadoux regime). Both are cells of one space.

co-located repeat surveys collapses SLI_ρ to $+0.006$; the grouped-leakage channel reports that 21.2% of test points leak through a shared location—numerically equal to $c_{\mathrm{obs}} = 0.212$, the two diagnostics agreeing—and `group_kfold()` drives it to 0%. Empirical optimism is a modest $+1.8\%$ (co-located records differ by year; weak signal), a textbook illustration of the leakage-vs-optimism distinction in the wild. The emulator *correctly refused* the query (n below its training envelope), demonstrating the area-of-applicability guard on real data and deferring to the empirical route.

8.7 Designing validation surveys

Finally, `design_validation()` inverts the SLI to place validation points. Over 120 replicates, estimating true map RMSE from a budget of 30 points, the optimal (Neyman, inclusion-weighted) design attains 45% **lower standard deviation** than random placement for the same budget, with negligible bias—i.e. the same precision from 30 points that random placement needs many more to reach. This unifies leakage diagnosis with spatial survey design.

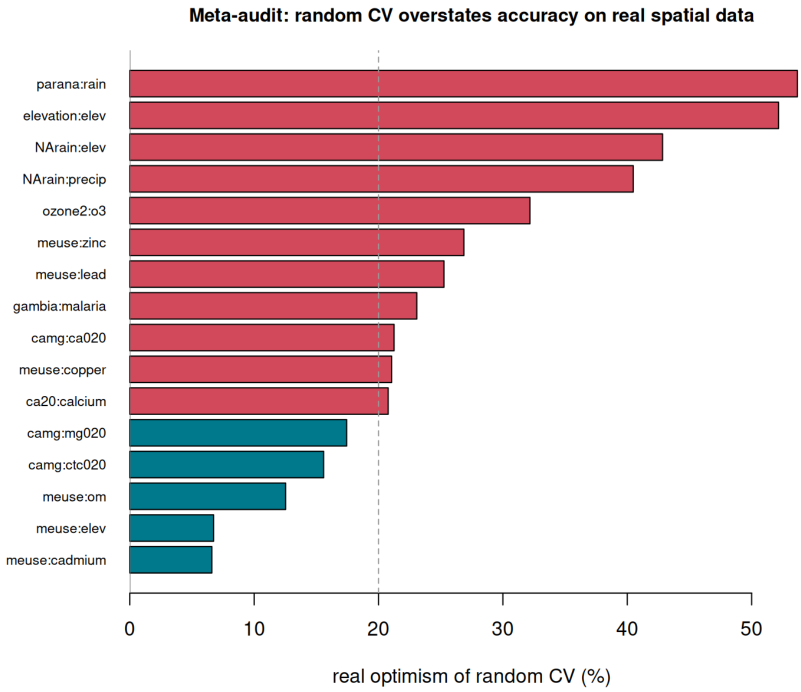

Figure 4: Meta-audit of 16 real spatial response variables. Naive random CV is optimistic for every dataset (median 22%, two-thirds > 20%). The trend-dominated fields (rainfall, elevation, ozone) show the largest optimism yet the smallest short-range leakage signal—the second, trend-extrapolation channel.

9 Discussion

What the framework settles. The Milà–Wadoux controversy is not a disagreement about facts but about regimes. Once design, estimand and target are declared, optimism is a well-posed, signed quantity, and the data decide which camp's prescription applies: clustered/convenience samples make random CV optimistic (Milà); probability samples make spatial CV pessimistic (Wadoux). The signed SLI and the optimism estimator report which regime a given analysis is in, and the recommender prescribes accordingly—conditionally, never neutrally-but- secretly-partisan.

Relationship to existing tools. `spLeakage` is complementary to, not competitive with, `CAST`, `blockCV`, `spatialsample` and `mlr3spatiotempcv`. They generate corrected folds; we audit an existing split, quantify its optimism, and—through `deleak_estimate()`—can even correct a published number without the original model. The reduction $|A| = W$ makes the bridge to NNDM explicit.

Scope and limitations. The closed-form theory is the Gaussian backbone: the single-NN approximation underestimates magnitude for dense training (corrected by a $\approx 1.6\times$ multi-neighbour factor or the exact $c_0^\top \Sigma^{-1} c_0$); non-Gaussian responses and sub-optimal learners (RF, GAM) are handled empirically by the emulator, for which the GP gives the envelope (real learners' explained variance is no larger than kriging's). The emulator is honest about its own extrapolation through the area-of-applicability guard, and abstains rather than guess—as it did on the Nigeria query. The most important open finding is the *second channel*: a large-scale trend-extrapolation component of optimism that all autocorrelation-based diagnostics (including ours) miss, now measurable via `trend_strength` but deserving a fully trend/covariate-aware optimism model. Finally, design-basedness must be elicited; no geometric test can certify a sample as a probability sample, so the recommender is decision *support*, not automation.

Outlook. The distribution-shift framing (7) points toward model-agnostic optimism bounds beyond the GP (importance weighting, integral-probability metrics), broadening the audience from spatial statistics to machine learning. Coupling the autocorrelation and trend channels into a single optimism model, and extending the emulator's generators to spatial GLMMs with covariate trends, are the natural next steps.

Conclusion. Reported accuracy for spatial maps is, across a wide range of real datasets, materially and systematically optimistic. `spLeakage` makes that optimism well-posed, measurable, signed, correctable and—through validation design—cheaper to pin down, while reconciling a controversy that has left practitioners without clear guidance. By auditing rather than generating folds, it adds a missing diagnostic layer to the spatial-prediction workflow.

Data and code availability

The `spLeakage` package, including all analysis scripts in `data-raw/` that reproduce the results in Section 8, is openly available at `https://github.com/olatunjijohnson/spLeakage` under an MIT licence. All datasets used are public (Malaria Atlas Project via the `malariaAtlas` R package; `meuse`, `ca20`, Paraná and the remaining meta-audit fields via `sp`, `geoR` and `gstat`).

Acknowledgements

The author thanks the developers of the spatial cross-validation ecosystem (`CAST`, `blockCV`, `spatialsample`, `mlr3spatiotempcv`) whose work this package builds upon and audits.

Conflict of interest

The author declares no conflict of interest.

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